Quantum Physics Simulations Using the PhET Interactive Platform

Esteban Salcedo
Department of Physics, LaGuardia Community College, City University of New York (CUNY)
esteban.salcedo@live.lagcc.cuny.edu

June 10, 2026

Abstract :

This study focuses on the use of simulation resources from PhET with the purpose of interacting with quantum phenomena. Four specific simulations were selected to study the hydrogen atom model by analyzing photon emissions and quantized energy levels, quantum measurements through coin-toss probability, semiconductor applications using a PN junction model, and quantum tunneling by attempting to calculate the wave function in different regions. The resulting visual comparisons illustrate the quantized nature of the atom as proposed by Niels Bohr, the fundamental role of probability in quantum mechanics, the importance of complex numbers in the Schrödinger equation, and the strong relationship between the transmission coefficient, the length of a potential barrier, and the total energy of an electron in quantum tunneling, together with the wide range of applications derived from this phenomenon.

Index Terms: Quantum mechanics, quantum tunneling, hydrogen atom, Bohr model, Schrödinger equation, PhET simulations, semiconductor physics, transmission coefficient, wave function, spectroscopy.

1 Introduction

In this section, the basic ideas behind quantum physics that are important for understanding the simulations used in this work are introduced. Key historical experiments and concepts, such as blackbody radiation, the photoelectric effect, and atomic spectra, are briefly reviewed, since they demonstrated the limitations of classical physics at microscopic scales. These developments led to the formulation of quantum theory, which describes the behavior of matter and energy at the atomic level.

1.1 *PhET Simulation Online Resources*

Physical Education Technology (PhET) is a project established in 2002 at the University of Colorado Boulder that provides an open-source platform designed to improve students' understanding through interactive computer simulations.[1]. This platform was designed in response to the challenging and abstract concepts that many students struggle to fully comprehend.

In physics, there are more than 100 different simulations focusing on a wide variety of physical phenomena. These simulations are animated and allow students to interact directly with the models and experiments. For physics majors, it is crucial to develop a strong understanding of the fundamental concepts, especially those described as threshold concepts, in order to avoid future misconceptions [2].

Part of these physics simulations focuses on quantum phenomena, offering an amplified representation of the microscopic world. Some of these simulations include quantum measurement, models of the hydrogen atom, blackbody spectrum, photoelectric effect, and quantum tunneling, among others. Students can manipulate different parameters such as photon energy, excited states, frequencies, and subatomic particle characteristics [2]. An example of the blackbody spectrum simulation is shown in Fig. 1.

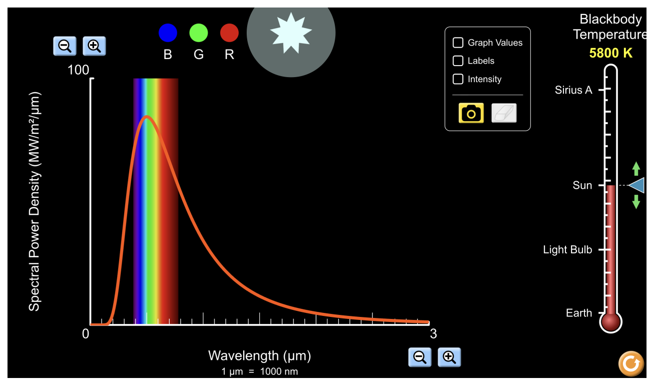

Figure 1. The blackbody spectrum simulation on the PhET platform allows the temperature to be adjusted as a key parameter, while the resulting spectral curves are plotted and recorded. Wavelength, power density, and intensity values are displayed in real time. [6]

1.2 *Quantum Physics Background*

The nature of light has been a topic of scientific interest for centuries, and the results obtained from a wide variety of research are abundant. A series of experiments that could not be explained by classical physics, such as the ultraviolet catastrophe and the photoelectric effect, provided evidence of a deeper physical mystery.

It was Max Planck's hypothesis of energy quantization that successfully explained blackbody radiation in agreement with the experimental curve [5]. This quantized energy hypothesis opened the door to the quantum world. Later, Albert Einstein followed Planck's idea to explain the photoelectric effect by introducing the concept of photons, which exhibit both particle-like and wave-like behavior.

These findings inspired many other scientists, and different types of experiments were carried out. One example is the Compton effect, which involves short-wavelength light (X-rays) directed at various materials. It was found that the scattered light emerges at different angles with a lower frequency, indicating a loss of energy [3]. Another experiment, that will later become the groundwork for Niels Bohr's contributions was the study of atomic spectra. When solids, liquids, or dense gases are heated, they emit light with a continuous spectrum due to strong interactions between atoms and molecules. In contrast, rarefied gases do not produce a continuous spectrum; instead, they emit light only at specific wavelengths, forming a discrete line spectrum.

In an attempt to explain the hydrogen emission spectrum, Niels Bohr combined the planetary model from classical mechanics with the quantum concept of photons. Taking into account Rutherford's experimental discovery of the positively charged atomic nucleus and introducing three postulates to overcome certain classical limitations, Bohr developed the first quantum model of the atom [5]. This work is regarded as one of the greatest achievements in quantum physics because it explained the discrete spectral lines of hydrogen and introduced the idea that electrons can occupy only specific quantized energy levels within the atom.

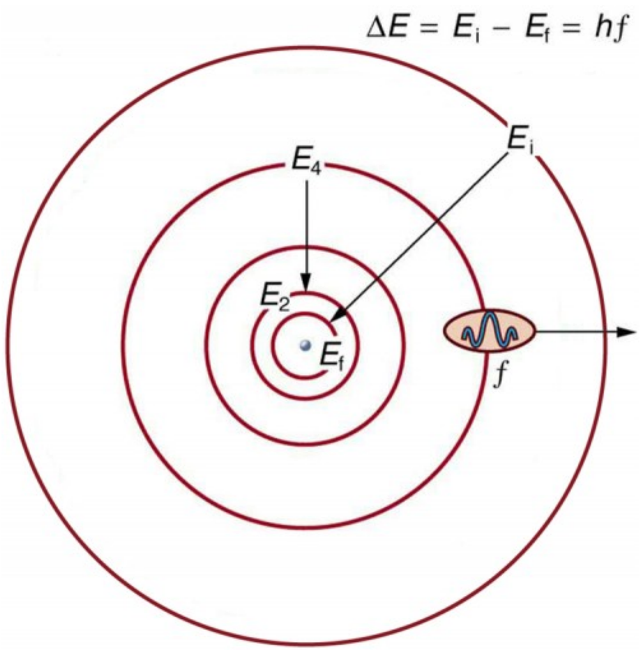

Figure 2. Bohr's model of the hydrogen atom proposes that electrons occupy discrete orbits around the nucleus and gain or lose energy through the absorption or emission of electromagnetic radiation.

This contribution was a major milestone for the scientific community. However, the Bohr model was not able to explain why the electron orbits were quantized until about a decade later, when Louis de Broglie proposed that all particles are subject to wave–particle duality [3]. Consequently, electrons can also behave as waves, and the quantization of the orbits can be understood as a result of standing-wave resonance conditions around the nucleus.

Bohr's model of the atom was an important approach despite its limitations, particularly its inability to accurately describe multi-electron atoms or predict the spectra of more complex elements such as helium [3]. However, scientists sought a more complete theoretical framework capable of providing more accurate predictions of quantum phenomena. This motivation led to the development of quantum mechanics.

2 Quantum Mechanics

Quantum mechanics successfully explains all the quantum phenomena mentioned previously, as well as the behavior of atoms and molecules. Moreover, quantum mechanics satisfies the correspondence principle, meaning that when applied to the macroscopic world, it approximates the predictions of Newtonian mechanics [3].

2.1 *The Schrödinger Equation*

Erwin Schrödinger published his famous wave equation in 1926 in an attempt to explain the propagation of matter waves. The background behind this equation is very abstract; however, it is possible to work with the equation without fully understanding its complete derivation. An analogy can be made with Newtonian equations, which are treated as fundamental relations upon which classical physics is built. Similarly, the Schrödinger equation serves as one of the fundamental equations of quantum mechanics[12].

$$i\hbar \frac{\partial \Psi(x,t)}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2 \Psi(x,t)}{\partial x^2} + V(x,t)\Psi(x,t) \tag{1}$$

Where $\Psi(x,t)$ represents the time dependant wave function, and $V(x,t)$ is the potential energy. If the potential V does not depend on time, it is possible to show that.

$$\Psi(x,t) = \psi(x)e^{i(kx-\omega t)} \tag{2}$$

Where $\psi(x)$ is a time-intependant function and $e^{-i\omega t}$ is space-independant function, meaning that the wave function is separable into two part: space-only part and time-only part[5], and this does not change the probability density results since $|\Psi|^2 = |\psi|^2$. In addition, $e^{-i\omega t}$ can also be express as $e^{-iEt/h}$ (the time modulation factor) according to de Broglie, the Energy of matter is given by $E = \hbar\omega$. Therefore, the equation 2 can be reduced to equation 3.

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x) \tag{3}$$

Equation 3 is known as the time-independent Schrödinger equation and is a very useful tool for cases such as the particle in a box, the harmonic oscillator, and quantum tunneling, in which the potential energy is time-independent.

2.2 *Quantum Tunneling in potential barriers*

Quantum tunneling is a phenomenon of high interest in quantum mechanics because it appears to violate the principles of classical mechanics [5]. An analogy for this situation is a ball with 100 J of energy rolling toward a 200 J hill. In classical mechanics—and in common sense—it is clear that the ball does not have enough energy to overcome the barrier and reach the other side. However, in quantum mechanics, due to the wave nature of particles, there is a chance that a particle can be found on the other side of a barrier. This occurs because the wave function is defined over all space.

In quantum mechanics, this situation can be describe by the following potential-energy function:

$$V(x) = \begin{cases} 0, & x < 0 \\ V_0, & 0 \le x \le a \\ 0, & x > a \end{cases} \tag{4}$$

Imagine a beam of electrons incident on a potential barrier. The probability that an individual particle will tunnel through the barrier can be determined by solving a boundary-value problem, as illustrated in Figure 3. Two conditions must be fulfilled: the wave

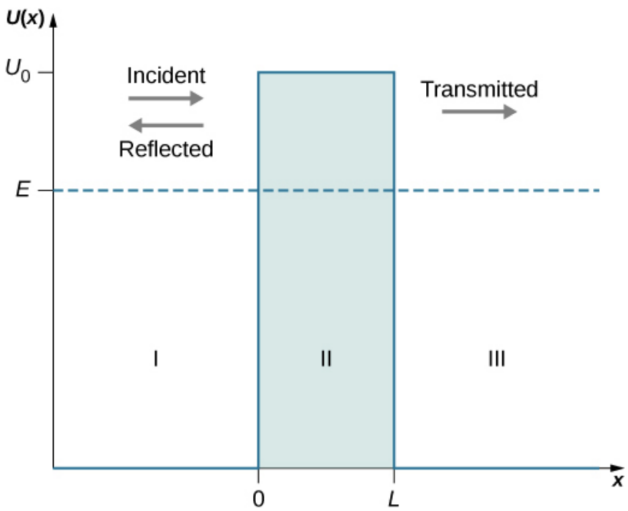

Figure 3. A potential barrier creates three regions: in region I the wave is partly reflected and partly incident, in region II part of the wave tunnels through the barrier, and in region III the transmitted wave behaves like a free particle. [5]

function ψ must be continuous at every point, and its first derivative must also be continuous for all x. The solution describing regions I, II, and III can be written as the sum of ψ_I, ψ_{II}, and ψ_{III}.

$$-\frac{\hbar^2}{2m}\frac{d^2\psi_{\mathrm{I}}(x)}{dx^2} = E\psi_{\mathrm{I}}(x), \text{ in region I: } -\infty < x < 0 \tag{5}$$

$$-\frac{\hbar^2}{2m}\frac{d^2\psi_{\mathrm{II}}(x)}{dx^2} + U_0\psi_{\mathrm{II}}(x) = E\psi_{\mathrm{II}}(x), \text{ in region II: } 0 \le x \le L \tag{6}$$

$$-\frac{\hbar^2}{2m}\frac{d^2\psi_{\mathrm{III}}(x)}{dx^2} = E\psi_{\mathrm{III}}(x), \text{ in region III: } L < x < +\infty \tag{7}$$

Equation 5 describes region I, where $V(x) = 0$ and the boundary condition $\psi_I(0) = \psi_{II}(0)$ must be satisfied. Moreover, region I contains two waves in a superposition state: the incident wave and the reflected wave [5]. In region III is being described by equation 7, $V(x) = 0$ and only the transmitted wave is present. The corresponding boundary condition is given by $\psi_{II}(L) = \psi_{III}(L)$. Therefore the solutions for ψ_I and ψ_{III} must be in the following way:

$$\psi_{\mathrm{I}}(x) = Ae^{ikx} + Be^{-ikx} \tag{8}$$

$$\psi_{\mathrm{III}}(x) = Fe^{ikx} \tag{9}$$

We can write the incident and reflected wave functions in region I as $\psi_{\mathrm{in}}(x) = Ae^{ikx}$ and $\psi_{\mathrm{ref}}(x) = Be^{-ikx}$, and $\psi_{tran}(x)$ for the transmitted wave in region III. The corresponding amplitude coefficients can be determined as follows:

$$|\psi_{in}|^2 = \psi^*_{in}(x)\psi_{in} = |A|^2 \tag{10}$$

$$|\psi_{ref}|^2 = \psi^*_{ref}(x)\psi_{ref} = |B|^2 \tag{11}$$

$$|\psi_{tran}|^2 = \psi^*_{tran}(x)\psi_{tran} = |F|^2 \tag{12}$$

The transmition probability is the ratio between $|F|^2$ and $|A|^2$ [5]

$$T(L,E) = \frac{|\psi_{\mathrm{tran}}(x)|^2}{|\psi_{\mathrm{in}}(x)|^2} = \left|\frac{F}{A}\right|^2 \tag{13}$$

Equation 6 can be rearrange as

$$\frac{d^2\psi_{\mathrm{II}}(x)}{dx^2} = \beta^2\psi_{\mathrm{II}}(x) \tag{14}$$

Where $\beta^2 = 2m/\hbar^2(U_0 - E)$, knowing that β is a positive number since $U_0 > E$. The solution for ψ_{II} is in the for of exponential oscillatory.

$$\psi_{\mathrm{II}} = Ce^{-\beta x} + De^{\beta x} \tag{15}$$

The ratio F/A is a parameter of interest [5], according to Eq. 10, and can be obtained (after a rigorous mathematical derivation [5]) as follows:

$$\frac{F}{A} = \frac{e^{-ikL}}{\cosh(\beta L) + i(\gamma/2)\sinh(\beta L)} \tag{16}$$

Where $\gamma \equiv \beta/k - k/\beta$ and after applying the boundary conditions to the previous equations and taking the complex conjugate of Eq. 16, the following relation is obtained:

$$T(L,E) = 16\frac{E}{V_0}\left(1 - \frac{E}{V_0}\right)e^{-2\beta L} \tag{17}$$

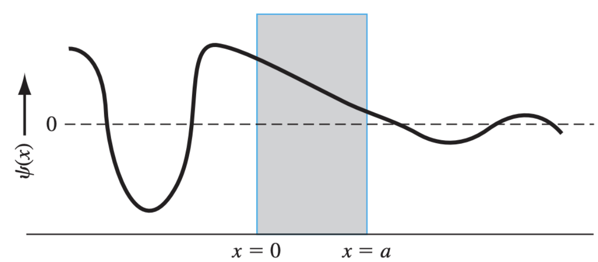

Figure 4. The wave function through the potential barriers [4]

These findings are extremely useful for many modern technological applications, such as **semiconductors**, since they belong to the field of nanotechnology [5], where quantum phenomena begin to manifest themselves more noticeably. In these devices, phenomena such as quantum tunneling occur very frequently, and this effect has become an important topic of interest with applications in many modern technologies and electronic devices.

3 Methods

Next, several PhET simulations will be explored in an attempt to visualize the physical phenomena previously analyzed. The simulations used in this section are Model of the Hydrogen Atom, Quantum Measurement, Semiconductor, and Quantum Tunneling. Interactions with these simulations will allow us to carry out measurements and observations, including the analysis of atomic spectra, the statistical behavior of quantum measurements, electron transport in a PN junction, and the probabilities of particles penetrating a potential barrier.

3.1 Model of the Hydrogen Atom

The goal of this simulation is to observe the quantized behavior of the spectrum obtained with the spectrometer provided by the simulation and to identify each energy-level transition within the different hydrogen spectral series using the corresponding equations discussed previously.

This simulation offers many adjustable settings, including different atomic models such as the Bohr, de Broglie, and Schrödinger models. The type of wavelength can also be adjusted to either white light (a combination of different wavelengths) or a specific monochromatic light.

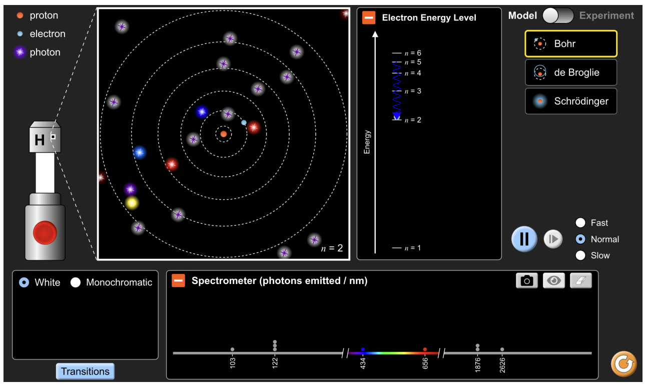

Figure 5. Interactive simulation of the Hydrogen Atom model on the PhET platform. Photons of different wavelengths travel through the atom, colliding with the electron and changing its quantum state [7].

In the right side there is a box displaying the energy levels and register every change of the electron when absorbs or emitts a photon. At the bottom, there is a box that records the wavelengths of the photons emitted when the electron transitions between energy levels. This first mode of the simulation allows the user to interact with all parameters and provides a clear visual representation of atomic behavior.

Once the user is familiar with the interface of the simulation, change to experimental mode by clicking on the corner right at the top. The whole setting will change to a mode in which only the spectrometer box is giving information.

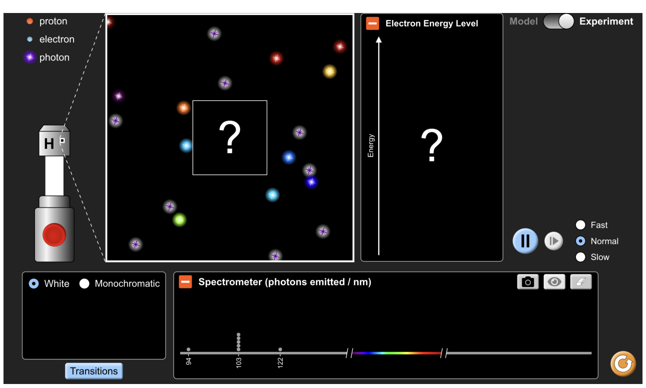

Figure 6. Experimental mode simulation model of the hydrogen atom [7].

The activity steps are as follows:

1. Set the simulation to experimental mode.
2. Select the white light source and turn it on.
3. Record the wavelength of the emitted photon using the spectrometer.
4. Calculate the frequency of the photon using:

$$f = \frac{c}{\lambda}$$

5. Determine the energy change of the electronic transition using:

$$\Delta E = hf$$

6. Identify the initial energy level n and the final energy level m associated with the transition.
7. Classify the transition according to the corresponding hydrogen spectral series:
    - Lyman series
    - Balmer series
    - Paschen series
    - Brackett series
    - Pfund series
8. Repeat the procedure for ten emitted photons.
9. Analyze the results to identify patterns in the hydrogen emission spectrum.

Taking Bohr's postulates into account together with the Rydberg formula for convenience.

$$\frac{1}{\lambda} = R\left(\frac{1}{m^2} - \frac{1}{n^2}\right) \tag{18}$$

3.2 Quantum Measurement

The objective of this simulation is to explore the statistical behavior of quantum measurements and compare it with classical results obtained from repeatedly tossing a coin over a large number of trials.

This simulation focuses on quantum superposition for different states. Flipping a coin, as simple as it may seem, can also be described from a quantum mechanics perspective. In classical physics, even though the coin flips in the air and lands on one side, there is never a true superposition of states. In quantum mechanics, however, the coin can exist in both states simultaneously (heads and tails) until a measurement is performed.

The simulation offers two versions of the coin toss experiment: the classical coin and the quantum coin, with the purpose of comparing their results. Each setting provides two different ways to perform the experiment. One consists of a single-coin measurement, while the other—required for this activity—consists of multiple-coin measurements.

$$|\psi\rangle = \alpha|H\rangle + \beta|T\rangle \tag{19}$$

Equation 19 describes the quantum superposition states of the simulation. The goal for this section is to compare the statistical behavior of a classical coin and a quantum coin.

1. Open the PhET Quantum Coin Toss simulation.
2. Select the classical coin mode.
3. Perform three sets of experiments:
    - 5 independent trials with $N = 5$ tosses each
    - 5 independent trials with $N = 100$ tosses each

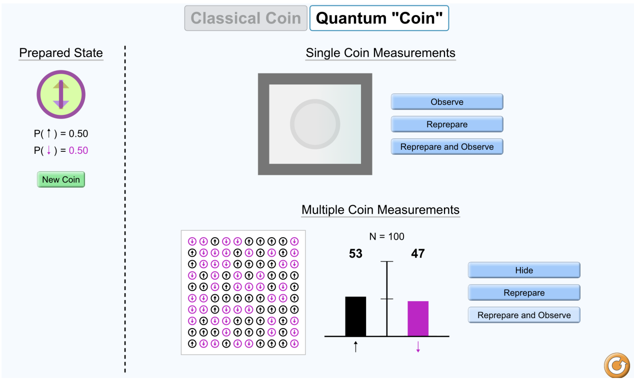

Figure 7. Quantum Tunneling of a Plane Wave through a Square Potential Barrier [8]

- 5 independent trials with $N = 10000$ tosses each

4. For each trial, record the number of heads and tails.
5. Repeat the same procedure using the quantum coin mode.
6. Compare the probability distributions for both classical and quantum cases.
7. Analyze how increasing the number of measurements affects the stability of the probabilities.
8. Observe how measurement influences the quantum coin state.

3.3 Semiconductor simulation

The objective of this simulation is to study how electrons move through a P–N junction and to determine the probability that an electron can tunnel through the depletion region at different applied voltages.

Figure 8 shows a simulation of a P-N junction, which is the basic building block of most modern electronics like diodes and transistors. This one shows two different semiconductors pushed together and :

- **The Left Side (N-type)**: Contains extra electrons (the blue spheres in the top conduction band).
- **The Right Side (P-type)**: Contains "holes" or missing electrons (represented by the blue spheres sitting in the lower valence band).

Determine the probability of an electron tunneling through the PN-junction depletion region under five different simulation conditions, corresponding to battery voltages of 0.2 V, 0.7 V, 1.0 V, 1.4 V, and 1.9 V. Assume a potential barrier height of $V_0 = 2$ eV and a depletion region width of 2 nm.

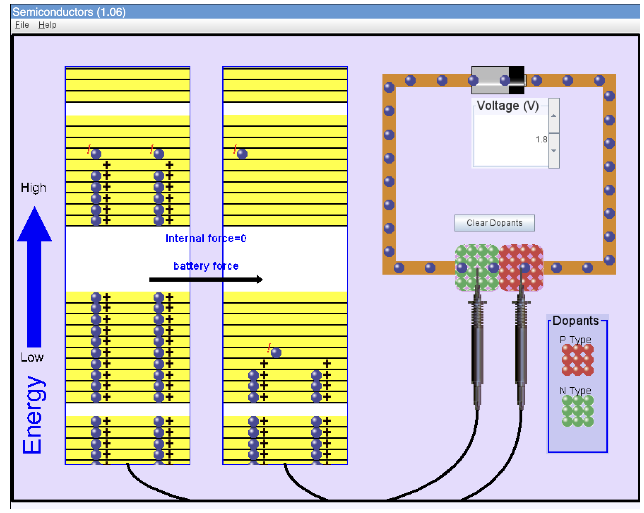

Figure 8. PhET Simulation: Electron and Hole Movement in a Semiconductor Diode [10]

3.4 Quantum Tunneling and Wave Packages

The aim of this simulation is to determine the amplitudes (A), (B), (C), (D), and (F) associated with the wave functions in the three regions of a quantum tunneling system. By analyzing the behavior of an electron incident on a potential barrier, the simulation provides the information needed to construct the wave functions in Regions I, II, and III and to study how their amplitudes are related through the boundary conditions imposed by quantum mechanics.

For this activity, it is necessary to use this simulation since it offers detailed and interactive graphs in which parameters such as total energy, potential energy, barrier length, wave direction, and wavefunction forms can be adjusted.

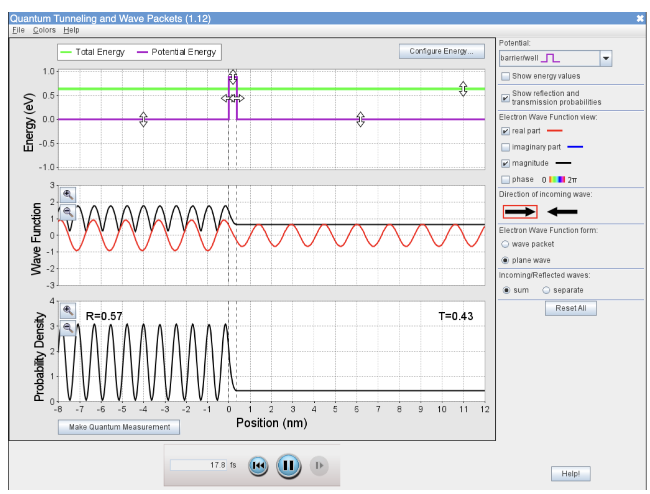

Figure 9. Quantum Tunneling of a Plane Wave through a Square Potential Barrier [9]

Before beginning the activity, it is necessary to configure the simulation settings. Click on the top-right corner where it says *Potential* and change it to *Barrier/Well*. Then, go to the *Electron Wave Function* settings and select the *Plane Wave* option. Finally, make sure the wave direction is set from *left to right*.

Calculating the wave function amplitudes: The goal of this section is to obtain the wave functions for regions I, II, and III previously discussed. Recall that region I consists of a superposition of the incident and reflected wave functions. From the simulation, record the transmitted and reflected coefficients.

The core idea is to **solve the system of equations** to find the values of A, B, C, D, and F, as explained in the previous section, making use of the four **boundary conditions.**

$$\psi_{\text{I}}(0) = \psi_{II}(0) \tag{20}$$

$$\psi_{\text{II}}(L) = \psi_{III}(L) \tag{21}$$

$$\left.\frac{d\psi_{\text{I}}(x)}{dx}\right|_{x=0} = \left.\frac{d\psi_{\text{II}}(x)}{dx}\right|_{x=0} \tag{22}$$

$$\left.\frac{d\psi_{\text{II}}(x)}{dx}\right|_{x=L} = \left.\frac{d\psi_{\text{III}}(x)}{dx}\right|_{x=L} \tag{23}$$

There are five unknowns and four equations. Therefore, Eq. 16 and Eq. 17 are used as additional constraints. **The coefficient F is assumed to be a real number** for exploratory modeling purposes, and its approximate value is estimated from the PhET simulation. MATLAB is highly recommended for solving the resulting system of equations due to its simplicity and computational efficiency. Use the example code as a guide for developing your numerical implementation.

Listing 1: MATLAB code used to solve the tunneling amplitudes.

```
% Physical constants
hbar = 1.0545718e-34;  %  Js
m = 9.10938356e-31;    % electron mass (kg)
e = 1.602176634e-19;   % electron charge (C)

% Input from simulation
L_nm = 0.2;            % barrier width in nm
L = L_nm * 1e-9;       % convert to meters
E_eV = 0.6;            % particle energy in eV
V0_eV = 1.0;           % barrier height in eV
F = 0.85;             % transmitted amplitude (from PhET)

% Convert energies to Joules
E = E_eV * e;
V0 = V0_eV * e;

% Wave numbers (physics-based)
k = sqrt(2*m*E) / hbar;
Beta = sqrt(2*m*(V0 - E)) / hbar;

% Gamma parameter
gamma = (Beta/k) - (k/Beta);

% Compute A from F/A relation
numerator = exp(-1i * k * L);
denominator = cosh(Beta * L) + 1i*(gamma/2)*sinh(Beta * L);
F_over_A = numerator / denominator;
A = F / F_over_A;

% Solve for B, C, D
syms B C D
eq1 = A + B == C + D;
eq2 = -1i*k*(A - B) == Beta*(D - C);
eq3 = C*exp(-Beta*L) + D*exp(Beta*L) == F*exp(1i*k*L);

sol = solve([eq1, eq2, eq3], [B, C, D]);

% Display results
fprintf('Results␣for␣amplitudes:\n');

fprintf('A␣=␣%.4f␣+␣%.4fi\n', real(double(A)), imag(double(A)));
fprintf('B␣=␣%.4f␣+␣%.4fi\n', real(double(sol.B)), imag(double(sol.B
     )));
fprintf('C␣=␣%.4f␣+␣%.4fi\n', real(double(sol.C)), imag(double(sol.C
     )));
fprintf('D␣=␣%.4f␣+␣%.4fi\n', real(double(sol.D)), imag(double(sol.D
     )));
fprintf('F␣=␣%.4f\n', F);
```

Five different settings are required while maintaining the same total energy and varying the width of the barrier **(0.2 nm, 0.4 nm, 0.6 nm, 0.8 nm, 1 nm)**. Record the tunneling probability for each case and analyze its behavior as the barrier length is changed.

4 Results

In this section, the results from the different PhET simulations are shown and discussed. The results include the hydrogen atom spectrum, coin toss experiments, and electron tunneling in a semiconductor. These results are used to compare theory with the simulations and to see how quantum effects appear in different situations.

4.1 Model of Hydrogen atom simulation

Table 1
Frequency of observed wavelengths from spectrometer simulation.

Wavelength (nm)	Occurrences
95	1
97	1
103	2
122	5
486	1
656	1
1876	1
2626	1
4052	1

Table 2
Hydrogen emission spectrum analysis. The wavelength of emitted photons is used to determine frequency, energy difference, quantum transitions ($m \rightarrow n$), and spectral series classification.

λ (nm)	Frequency (Hz)	ΔE (eV)	m	n	Type of Series
95	3.16×10^{15}	13.06	1	4	Lyman series
97	3.09×10^{15}	12.78	1	4	Lyman series
103	2.91×10^{15}	12.04	1	3	Lyman series
122	2.46×10^{15}	10.16	1	2	Lyman series
486	6.17×10^{14}	2.55	2	4	Balmer series
656	4.57×10^{14}	1.89	2	3	Balmer series
1876	1.60×10^{14}	0.66	3	4	Paschen series
2626	1.14×10^{14}	0.47	4	5	Brackett series
4052	7.40×10^{13}	0.31	5	6	Pfund series

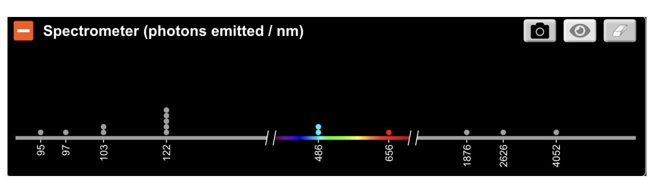

Figure 10. Discrete Emission Spectrum of Atomic Hydrogen from PhET Simulation [7]

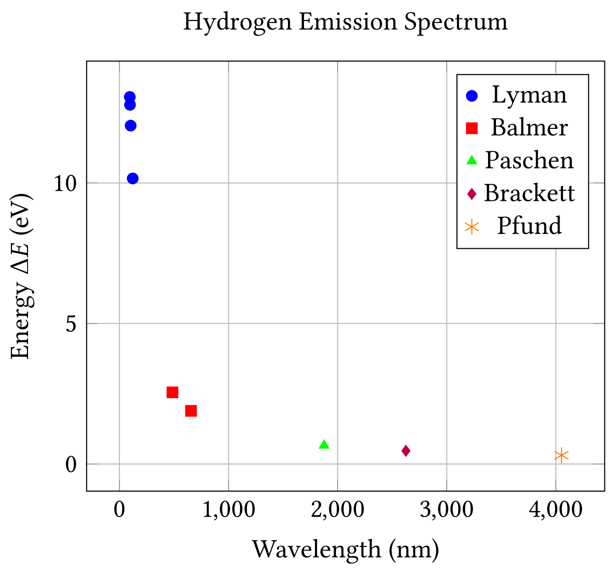

Figure 11. Hydrogen emission spectrum showing the separation of spectral series and their corresponding emitted photon energies.

4.2 Quantum Measuremnt Coin toss

Table 3
Classical coin toss results with probabilistic analysis. Each value of N corresponds to five independent trials.

N	Heads	Tails	$P(H)$	$P(T)$
10	5	5	0.50	0.50
10	5	5	0.50	0.50
10	5	5	0.50	0.50
10	3	7	0.30	0.70
10	2	8	0.20	0.80
100	42	58	0.42	0.58
100	47	53	0.47	0.53
100	46	54	0.46	0.54
100	42	58	0.42	0.58
100	50	50	0.50	0.50
1000	4988	5012	0.4988	0.5012
1000	4943	5057	0.4943	0.5057
1000	4948	5052	0.4948	0.5052
1000	4964	5036	0.4964	0.5036
1000	4998	5002	0.4998	0.5002

Table 4
Quantum coin measurement results with probabilistic analysis. Each value of N corresponds to five independent trials.

N	Heads	Tails	$P(H)$	$P(T)$
10	2	8	0.20	0.80
10	4	6	0.40	0.60
10	4	6	0.40	0.60
10	4	6	0.40	0.60
10	10	0	1.00	0.00
100	44	56	0.44	0.56
100	47	53	0.47	0.53
100	53	47	0.53	0.47
100	45	55	0.45	0.55
100	49	51	0.49	0.51
10000	4951	5049	0.4951	0.5049
10000	4920	5080	0.4920	0.5080
10000	4976	5024	0.4976	0.5024
10000	5014	4986	0.5014	0.4986
10000	5048	4952	0.5048	0.4952

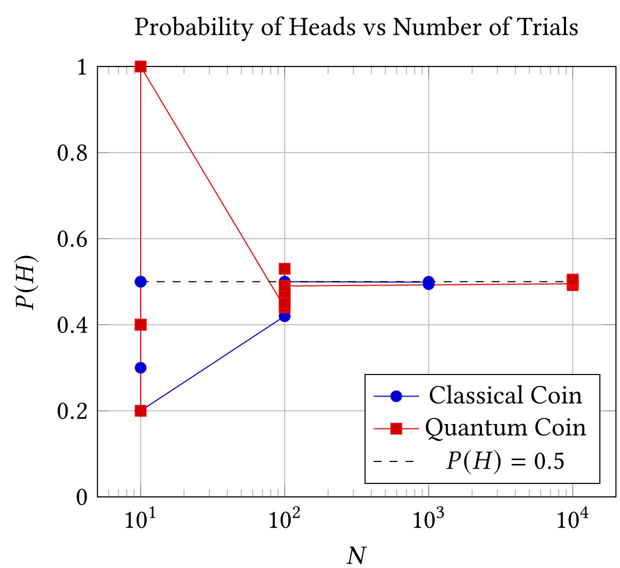

Figure 12. Comparison of classical and quantum coin probabilities showing convergence to the theoretical value $P(H) = 0.5$ as N increases.

4.3 Simulation of the Transmission Coefficient in a Semiconductor

Table 5
Transmission coefficient through a PN-junction depletion barrier using $V_0 = 2$ eV and $L = 2$ nm. Electron energy is taken as $E = V$ in eV units.

Electron Energy E (eV)	E/V_0	β (m^{-1})	T
0.2	0.10	7.24×10^9	1.8×10^{-10}
0.7	0.35	5.70×10^9	2.9×10^{-8}
1.0	0.50	4.96×10^9	2.1×10^{-6}
1.4	0.70	3.41×10^9	3.6×10^{-5}
1.9	0.95	1.15×10^9	1.2×10^{-2}

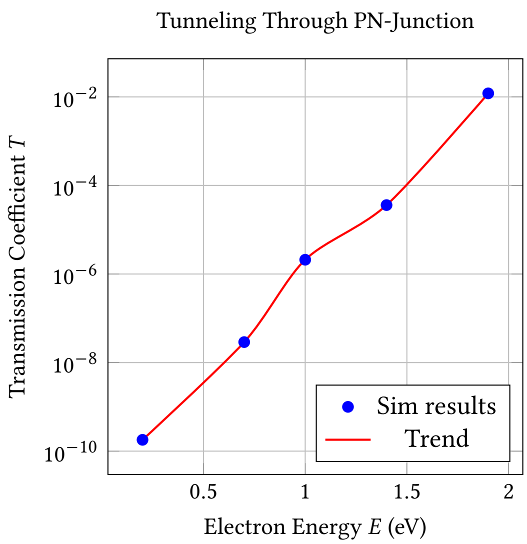

Figure 13. Transmission coefficient T vs energy E. The graph illustrates a directly proportional relationship between the transmission coefficient and the particle energy.

4.4 *Tunneling Wave Function Amplitude Calculations*

Table 6
Incident and reflected wave amplitudes for five barrier configurations.

Setting	A	B
Setting 1	0.8115 + 0.6533i	-0.3299 + 0.6418i
Setting 2	0.1875 + 1.1818i	-1.0202 + 0.3967i
Setting 3	-0.6304 + 0.8891i	-0.9757 - 0.4252i
Setting 4	-1.0812 + 0.1812i	-0.3926 - 1.0161i
Setting 5	-0.8461 - 0.6118i	0.4291 - 0.9498i

Table 7
Wavefunction amplitudes inside and after the barrier for five configurations.

Setting	C	D	F
Setting 1	0.2338 + 1.3466i	0.2478 - 0.0514i	0.85
Setting 2	-0.8971 + 1.5288i	0.0644 + 0.0497i	0.60
Setting 3	-1.6079 + 0.4434i	0.0018 + 0.0205i	0.30
Setting 4	-1.4701 - 0.8391i	-0.0037 + 0.0043i	0.16
Setting 5	-0.4155 - 1.5616i	-0.0015 + 0.0001i	0.08

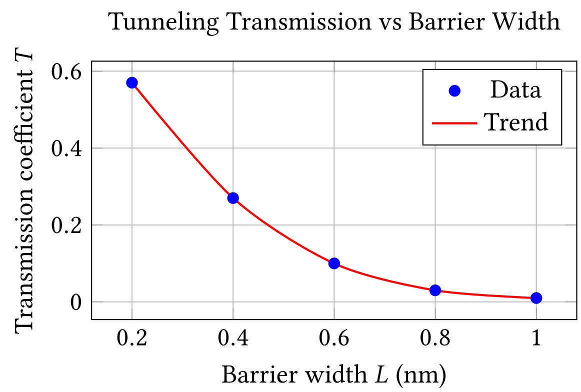

Figure 14. Transmission coefficient as a function of barrier width. The transmission coefficient T decays as the barrier length L increases.

5 Discussion

The Model of Hydrogen atom simulation illustrates the Bohr hypothesis and also allow us to interact with it. The animation makes it easier to visualize the planetary model with quantized orbits in action, while photons of different wavelengths continuously interact with the electron as it transitions from one energy state to another.

Figure 10 and Figure 11 show that the hydrogen atom emits photons of only specific wavelengths, reinforcing the idea of quantization on which the entire hypothesis is based.

From the **Quantum Measurement simulation** section, Figure 12 illustrates a comparison between the classical coin and the quantum coin. The visual shows that, although the results appear random for small sample sizes N, as the number of trials increases, both cases converge toward a probabilistic distribution, which is the fundamental statistical approach used in quantum mechanics. However, it is very important to emphasize the superposition condition of the quantum coin, in which the coin simultaneously exists in both states—heads and tails—until a measurement is made.

From the **Quantum Tunneling simulation**, Table 1 and Table 2 show the values of the amplitudes. As mentioned previously, F was assumed to be a real number, while the other amplitudes are complex quantities. This result makes it possible to rewrite the wave functions that describe each region of the system. As established before, this activity was merely exploratory, aimed at gaining familiarity with the wave function in the different regions; however, for a deeper analysis, a higher level of mathematical rigor is required.

Figure 14 illustrates the strong relationship between the barrier width and the transmission coefficient, showing an inversely proportional behavior. The larger the barrier width, the smaller the transmission coefficient becomes. In contrast, Figure 13, from the **semiconductor simulation**, shows the influence of the electron's total energy, since it is directly proportional to the transmission coefficient. This simulations describes how the electricity is being control inside the semiconductor. The PN junction illustrates the depletion region, where free electrons can tunnel through the barrier and recombine with empty holes on the other side. Figure 13 shows a small transmission coefficient. Although it may appear to represent a very low tunneling probability, it becomes meaningful when considering the large number of electrons incident on the barrier, many of which still successfully tunnel through and reach the other side.

6 Conclusion

The theory explains the three models of the hydrogen atom—Bohr, de Broglie, and Schrödinger. Simulations based on these models are extremely useful for educational purposes. It can take time to assimilate the concepts behind these theories; however, having visual and animated resources simplifies the process of acquiring this information. Students, regardless of their educational level, can easily comprehend the basic principles of these three models.

Spectroscopy is a revolutionary technique that allows us to study matter from planets, stars, and galaxies by analyzing the emission of light. Since every element has its own emission pattern, it acts like a fingerprint.

Quantum tunneling is a phenomenon that challenges our intuition, since human perception is grounded in the macroscopic world. However, this effect is of great interest to the scientific community due to its wide range of applications, including tunnel diodes, scanning tunneling microscopes, nuclear fusion in stars, flash memory, and quantum computing, and is also becoming increasingly known to a broader audience, particularly following the 2025 Nobel Prize in Physics awarded to John Clarke, Michel H. Devoret, and John M. Martinis *for the discovery of macroscopic quantum mechanical tunneling and energy quantization in an electric circuit* [11].

Semiconductors are a very broad field, and for our purposes regarding quantum tunneling, they provide a practical example of how quantum mechanical effects can be applied in modern technology, since they are designed on the scale of micro- and nanometers, where the laws of quantum mechanics become more noticeable.

Simulations such as PhET are a very useful resource for grasping the basic, abstract concepts of quantum mechanics. They allow us to interact directly with theoretical atoms or adjust parameters to analyze the resulting tunneling probability. This ability to visualize physical phenomena makes these concepts more accessible to a general audience.

References

[1] N. R. Alsalhi et al. "The Effect of Using PhET Interactive Simulations on Academic Achievement of Physics Students in Higher Education Institutions". In: *Educational Sciences: Theory & Practice* 24.1 (2024), pp. 65–75. DOI: 10.12738/jestp.2024.1.006.

[2] T. M. Clark and J. M. Chamberlain. "Use of a PhET Interactive Simulation in General Chemistry Laboratory: Models of the Hydrogen Atom". In: *Journal of Chemical Education* 91.8 (2014), pp. 1198–1202. DOI: 10.1021/ed400454p.

[3] Douglas C. Giancoli. *Physics: Principles with Applications*. 7th. Boston, MA: Pearson Education, 2014. ISBN: 978-0321625922.

[4] Donald A. Neamen. *Semiconductor Physics and Devices: Basic Principles*. 4th. New York: McGraw-Hill, 2012.

[5] OpenStax. *University Physics Volume 3*. OpenStax, Rice University, 2016. URL: https://openstax.org/details/books/university-physics-volume-3.

[6] PhET Interactive Simulations. *Blackbody Spectrum*. Accessed: 2026-05-09. 2025. URL: https://phet.colorado.edu/.

[7] PhET Interactive Simulations. *Models of the Hydrogen Atom*. https://phet.colorado.edu/en/simulations/models-of-the-hydrogen-atom. University of Colorado Boulder, Accessed: 2026-05-15. n.d.

[8] PhET Interactive Simulations. *Quantum Measurement*. https://phet.colorado.edu/en/simulations/quantum-measurement. University of Colorado Boulder, Accessed: 2026-05-15. n.d.

[9] PhET Interactive Simulations. *Quantum Tunneling and Wave Packets*. https://phet.colorado.edu/en/simulations/quantum-tunneling. University of Colorado Boulder, Accessed: 2026-05-15. n.d.

[10] PhET Interactive Simulations. *Semiconductors (Diodes, Transistors)*. https://phet.colorado.edu/en/simulations/semiconductor. University of Colorado Boulder, Accessed: 2026-05-15. n.d.

[11] The Nobel Prize. *The Nobel Prize in Physics 2025 – Press Release*. https://www.nobelprize.org/prizes/physics/2025/press-release/. Accessed: 2026-05-15. 2025.

[12] Paul A. Tipler and Ralph A. Llewellyn. *Modern Physics*. 5th. New York: W. H. Freeman and Company, 2008.